Assignment 4
Structured Products

1. Case study: Certificate Pricing.

On the 15th of February 2008 at 10:45 C.E.T., the Bank XX issues a certificate whose hedging termsheet is described in the annex.
Determine the upfront X% in the Certificate issue, in a single-curve interest rate modeling setting, knowing that the two counterparties have signed an ISDA with CSA that allows neglecting the counterparty risk. Neglect also the dynamics of interest rates.

Market parameters:
Basket underlyings: ENI $S_0 = 12.3 \ Euro$, AXA $S_0 = 22.1 \ Euro$
Correlation = 49.0%, $\sigma_1 = 20.1\%$, $\sigma_2 = 18.3\%$,
Dividend Yields: $d_E = 3.2\%$, $d_{CS} = 2.9\%$
Discounts: consider values of the 15th of February 2008 at 10:45 C.E.T.

2. Exercise: Pricing Digital option

Verify the difference between the price of a digital option computed according to the Black model and in the case where one takes into account the smile in the curve of the implied volatility.
Dataset:
- Volatility & smile.
- ATM forward (Reference in dataset is ATM Spot)
- Notional 10 Mln €
- Digital Payoff 5% Notional
- Expiry: 1y (Act/365)
- Discount curve of the 15th of February 2008 at 10:45 C.E.T.

3. Exercise: [Facultative] Pricing

Consider the model with the characteristic function

$$\phi^c(u) = \frac{1}{\left(1 - i\frac{u}{p^+}\right)\left(1 + i\frac{u}{p^-}\right)} e^{i\,\mu\,u}$$

where $p^+, p^- \in \Re^+$ with $p^+ > 1$ and μ is established according to the martingale condition for the corresponding exponential additive.

Compute the call price C (discount factor and ATM forward as above) with the Lewis formula for moneyness x=-5.223%, 0, +15%; $p^+ = 1.5, p^- = 0.9$.

a. Compute C with quadrature.
b. Compute C with the residuals' technique.
c. Compute C with MC. Is it possible to compute the CDF?
d. Compute C via FFT. Choose an adequate value for M and, and either ξ_1 or dz. Discuss the results for different choices of the FFT parameters.

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Remark: structure the code with i) a script: runPricingFourier, ii) some functions that compute I (with the different methodologies: e.g. FFT & Quadrature) that call iii) an integrand function. It is suggested not to use the parameters in the excel spreadsheet but to deduct them within the code (in order to avoid rounding errors).

4. Exercise: Pricing

Compute call prices C on the 15th of February 2008 at 10:45 C.E.T., according to a normal mean-variance mixture with $\alpha = 1/2$ (NIG) model with parameters $\sigma = 20\%; \kappa = 1; \eta = 3;$ t=1; for moneyness x between -25% and 25% (in a grid with 1% steps). F_0, discount and ttm are the values in the Exercise 2. Compute prices according to the following methods:

a. FFT. Choose an adequate value for M and, and either ξ_1 or dz. Discuss the results for different choices of the FFT parameters;
b. Quadrature;
c. MonteCarlo (simulating IG and Gaussian).
d. Facultative: Consider a α=1/3 model with same parameters (same values for a different model). Compute C with FFT and quadrature. Comment the results: Do you observe significant differences with respect to the above case in terms of precision of results?

Remark: As above, structure the code with i) a script: runPricingFourier, ii) some functions that compute I (with the two methodologies: FFT & Quadrature) that call iii) an integrand function. It is suggested not to use the parameters in the excel spreadsheet but to deduct them within the code (in order to avoid rounding errors).
Hint: The parameters of matlab IG simulation should be reconciled with the ones in the assignment e.g. using the Laplace transform of the corresponding mixture r.v. Compute numerically the first four moments and verify with the analytical ones.

5. Case study: Volatility surface calibration

Calibrate a normal mean-variance mixture with α=2/3 model parameters considering SP 500 volatility surface via a global calibration with constant weights. F_0, discount and ttm are the values in the Exercise 2.
Plot implied volatility obtained with the model and compare it with market's one.

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Exercise 4.1, Annex:

Indicative Terms and Conditions as of 15th of February 2008

Swap Termsheet

Principal Amount (N):	100 MIO EUR
Party A:	Bank XX
Party B:	I.B.
Trade date:	today
Start Date:	19 Feb 2008
Maturity Date (t):	5 years after the Start Date, subject to the Following Business Day Convention.

Party A pays:	Euribor 3m + 1.30%
Party A payment dates:	Quarterly, subject to Modified Following Business Convention
Daycount:	Act/360
Party A pays @ Maturity Date:	$(1-P)$ of the Principal Amount
Protection (P):	95%

Party B pays @ Start Date:	X% of the Principal Amount
Party B pays @ Maturity Date:	Coupon
Participation coefficient (α) :	110%
Coupon:	Pays at expiry date the participation coefficient of the performance, if positive, of an equally weighted basket of ENI S.p.A. and AXA S.A.

$$\alpha \cdot \left(S(t)-P\right)^{+}$$

Basket:

$$S(t)=\sum_{n=1}^{2}\frac{E_t^n}{E_0^n}\times W_n$$

and:

E_t^n : Value of the n^{th} element of the basket at Maturity Date t ;

E_0^n : Value of the n^{th} element of the basket at Start Date;

W_n : Weight of the n^{th} element of the basket, each one equal to one half..